Subani Shaik

Aws DevOps Automation Engineer

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PROFESSIONAL SUMMARY:

- Over **7+ Years** of experience in IT industry comprising of **Aws Engineer** and as a **DevOps Cloud Automation engineer**, including Software Configuration Management **(SCM)**, **Build/Release Management**, **Continuous Integration** and **Continuous Delivery** using different tools like **Jenkins, Terraform, Docker, Maven, Kubernetes**.
- Worked under the agile operation process in **DevOps team** (Build & Release automation, unit test automation, and Code review, Service, Incident and Change Management Environment,) to deliver an end-to-end continuous integration/continuous delivery product in an open-source environment using various tools like **Jenkins, Git CICD**.
- Hands-on experience with **AWS Lambda** workflow implementation using **python** to interact with application deployed on **EC2** instance and **S3 bucket**.
- Extensive expertise in AWS cloud services, including **EC2, EBS, EFS, Auto Scaling, Load Balancers, S3, VPC, IAM, CloudWatch**, and **Lambda**, with a proven ability to optimize cloud resources for cost-efficiency and performance.
- Proficient in Infrastructure as Code (IaC) using **Terraform** and **AWS CloudFormation** to automate and manage cloud infrastructure, ensuring consistency and repeatability across environments.
- Strong experience in building and maintaining **CI/CD pipelines** using tools like **Jenkins, GitLab CI/CD**, and **AWS CodePipeline, Aws Codebuild, Aws Code Deploy** to enable seamless software delivery and continuous integration.
- Skilled in containerization using **Docker** and orchestration with **Kubernetes** to deploy, scale, and manage containerized applications in production environments.
- Hands-on experience with monitoring and logging tools such as **CloudWatch, Prometheus, Grafana**, and **ELK Stack** to ensure system reliability, performance, and proactive issue resolution.
- Proficient in automation using scripting languages like **Python, PowerShell, Bash, and Ruby**, and configuration management tools like **Ansible, Chef, and Puppet** to streamline operational tasks and improve efficiency.
- Extensive knowledge of **Linux/Unix** administration, including system configuration, troubleshooting, performance tuning, and managing web servers like **Apache Tomcat, JBoss, and Nginx**.
- Strong background in cloud security practices, including implementing **IAM policies**, Role-Based Access Control (**RBAC**), and securing network configurations (**VPC, subnets, security groups**).
- Experience in multi-cloud environments (**AWS, Azure, GCP**) and managing resources across platforms to ensure flexibility and scalability.
- Proven ability to work in cross-functional teams and deliver results in fast-paced, **agile environments,** with excellent communication and collaboration skills.
- Adept at **incident response, troubleshooting**, and optimizing system performance for mission-critical applications, ensuring minimal downtime and maximum reliability.
- **Certified AWS Solution Architect, Aws Cloud Practioner and Python for Data Science, AI** with hands-on experience in deploying and managing cloud-native applications and infrastructure.
- Strong analytical and problem-solving skills, with a track record of delivering efficient and reliable technological solutions to complex challenges.
- Experience in automating routine tasks using **AWS CLI and SDKs**, and implementing infrastructure lifecycle management with tools like **Terraform**.

- Skilled in designing and configuring networking components (**VPC, subnets, security groups**) to establish secure and well-connected cloud environments.
- Created **Terraform modules** to encapsulate infrastructure components, promoting code reusability and maintainability across different projects.
- Managed infrastructure lifecycle with **Terraform**, **handling resource creation**, modification, and deletion in a controlled and auditable manner.
- **WebAPI** often follows **REST** (Representational State Transfer) principles, which emphasize a stateless client-server architecture, uniform interface, and resource-based interactions.
- Utilized **Terraform workspaces** to manage multiple environments (e.g., **development, staging, production**) and maintain separate state files for each, enhancing isolation and manageability.

TECHNICAL SKILLS:

Cloud Platforms	AWS (EC2, S3, RDS, Lambda, VPC, IAM, CloudFormation, CloudWatch, Route 53, ECS, EKS, Elastic Beanstalk, SNS, SQS, DynamoDB, API Gateway, CodePipeline, CodeBuild, CodeDeploy, ECR, EFS), Azure(VMs, Vnet, Azure Functions), GCP(GC Engine, VPC,GC Functions)
DevOps Tools	Jenkins, GitLab CI/CD, GitHub Actions, Terraform, Ansible, Puppet, Chef, Docker, Kubernetes, Helm, Prometheus, Grafana, ELK Stack, Nagios, Splunk
CI/CD Pipelines	Jenkins, GitLab CI/CD, AWS CodePipeline, GitHub Actions, CircleCI
Containerization	Docker, Kubernetes, ECS, EKS
Infrastructure as Code	Terraform, AWS CloudFormation, Ansible
Scripting Languages	Python, Powershell, Bash, Shell Scripting
Version Control	Git, GitHub, GitLab, Bitbucket
Monitoring & Logging	AWS CloudWatch, Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Splunk, Newrelic, Datadog
Operating Systems	Linux (Ubuntu, CentOS, RedHat), Windows Server
Databases	MySQL, PostgreSQL, MongoDB, DynamoDB, RDS
Networking	TCP/IP, DNS, VPN, VPC, Subnets, Route 53, Load Balancers (ALB, NLB)
Security	IAM, KMS, Secrets Manager, Security Groups, NACLs, AWS WAF, AWS Shield
Ticketing Tools	Service Now, Jira
Testing Tools	Selenium, Selenium Grid
Build Tools	ANT, Maven, Gradle
Web Servers	Apache Tomcat 8.x, Nginx (Forward/Reverse Proxy),
Serverless Tools	AWS Lambda, API Gateway, DynamoDB Streams, AWS Step Functions, EventBridge, SQS, SNS.

EDUCATION

- University of Southern Mississippi, MS in Computer Science
- University of Acharya Nagarjuna, BS in Computer Science

PROFESSIONAL CERTIFICATIONS / TRAINING

- AWS Certified Cloud Practioner , Aws Solution Architect ,AWS Partner: Sales Accreditation (Business)
- Certified in Python for Data Science, AI & Development

PROFESSIONAL EXPERIENCE:

Role: Aws DevOps Engineer **Feb 2024 – Till Date**

Client: Bayer Crop Science, Creve Coeur, MO ,USA

Responsibilities:

- Designed and implemented CI/CD pipelines using **Jenkins** and **AWS CodePipeline** to automate the build, test, and deployment processes for **microservices** and serverless applications such as **AWS Lambda**, ensuring faster and error-free deployments.
- Managed AWS infrastructure as code (IaC) using **Terraform** and **CloudFormation** to provision and maintain scalable, secure, and cost-effective cloud environments, reducing manual intervention and ensuring consistency.
- Containerized applications using **Docker** and orchestrated deployments with **Kubernetes (EKS)** to enable efficient scaling, management, and high availability of containerized workloads across multiple environments.
- Automated configuration management using **Ansible** to maintain consistent and reproducible environments across **development, staging,** and **production**, minimizing configuration drift and improving reliability.
- Implemented monitoring and alerting solutions using **CloudWatch, Prometheus,** and **Grafana** to proactively monitor application performance, detect anomalies, and ensure high availability of critical systems.
- Secured AWS environments by configuring **IAM roles, Security Groups, VPCs, NACLs**, and **KMS encryption** to enforce least-privilege access, network isolation, and data protection in compliance with security best practices.
- Developed and maintained Infrastructure as Code (IaC) scripts using **Terraform** and **PowerShell** to automate the provisioning of **virtual machines** (VMs) and cloud resources across **AWS, Azure**, and **GCP** ensuring rapid and consistent infrastructure deployment.
- Integrated DevOps tools such as **Docker, Kubernetes**, and **Jenkins** into the development workflow to streamline deployment processes, improve deployment speed, and enhance overall system reliability.
- Designed and developed a user-friendly GUI using **ReactJS, HTML**, and **Python** to enable clients to deploy virtual machines with pre-configured applications like **Zabbix, Graylog**, and **Trendmicro** in a single click, simplifying complex workflows.
- Conducted root cause analysis and resolved complex automation issues by optimizing **Terraform** and **PowerShell** scripts, ensuring robust, scalable, and maintainable cloud infrastructure.
- Automated the creation of **Virtual Machines** (VMs) with built-in applications (e.g**., Zabbix, Graylog, Trendmicro**) on **AWS, Azure**, and **GCP** using **Terraform** for provisioning and **Ansible** for post-provisioning configuration, reducing manual effort and improving efficiency.
- Collaborated with cross-functional teams, including **Servicenow, Ipaas**, and **Stackstorm**, to configure pipelines for passing parameter values to the **Remote Action Server** (**RAS**), ensuring seamless integration and automation.
- Supported VM provisioning by filling appropriate data in **ServiceNow**, running backend code to fetch client environment data, and publishing updated **AMIs** (**Golden Images**) to ensure secure and up-to-date VM deployments.
- Executed **Terraform** and **Ansible** commands on the RAS server by fetching credentials from **Hashicorp Vault** and code from **Bitbucket**, ensuring secure and automated provisioning and configuration of cloud resources.
- Collaborated with region leads and pod leads to gather scenarios for User Acceptance Testing **(UAT)**, ensuring successful **VM creation** and **post-provisioning steps**, and obtaining UAT sign-off from stakeholders.

- Developed an intuitive GUI using **ReactJS, HTML**, and **Python** to simplify VM deployment with pre-configured applications, enabling clients to deploy VMs with a single click and reducing dependency on technical expertise.
- Enforce cloud security best practices by defining and managing **IAM roles/groups**, **S3 bucket** policies, and conducting regular vulnerability assessments and penetration testing.
- Optimize cloud costs by leveraging **AWS Cost Explorer**, **Trusted Advisor**, and implementing cost-saving measures such as reserved instances, spot instances, and auto-scaling policies.
- Collaborate with cross-functional teams, including the ServiceNow team for **ITSM integration**, the **HashiCorp** team for **Terraform** and Vault implementations, and the **Full Stack team** for end-to-end application development and deployment.
- Provide rotational production support, resolving **P1 (critical), P2 (high),** and **P3 (medium)** incidents promptly, conducting root cause analysis, and implementing preventive measures to minimize downtime and ensure system reliability.
- Design and implement disaster recovery and high availability strategies using **AWS Backup, Route 53**, and multi-region redundancy to ensure business continuity and data resilience.
- Collaborate with the Full Stack team to ensure seamless integration of front-end and back-end components, enabling smooth deployment and operation of **eCommerce** and retail applications.
- Container builds/deployments using Multi-Stage Docker files, **Docker Compose** and **Kubernetes** stacks.
- Managed AWS infrastructure as code (IaaS) using **Terraform**. Expertise in writing new **python scripts** in order to support new functionality in **Terraform**. Provisioned the highly available **EC2 Instances** using **Terraform** and cloud formation and setting up the build and deployment automation for Terraform scripts using **Jenkins**
- Configured alarms in **CloudWatch** for monitoring the server's performance, health, **CPU utilization**, Memory and disk usage of AWS resources using **AWS CloudWatch**.
- Worked with container-based deployments using **Docker, Docker images**, **Docker file, Docker Compose** and **ECR** Docker registries for task and service definitions to deploy tasks on **AWS ECS** clusters on **AWS EC2** instances.
- Developed **Jenkins pipeline** to create the images of all the successful builds and to push them in to **AWS ECR** that is later consumed by the **Kubernetes (EKS)** cluster to do the rolling deployments by consuming the images.
- Implemented automated alerting and incident management using tools like **AWS CloudWatch Alarms**, resulting in a **20% reduction** in mean time to resolution **(MTTR)** for critical incidents.
- Collaborated with teams to design and deploy effective log management and analysis solutions using **ELK Stack** (**Elasticsearch, Logstash,** and **Kibana**) for improved troubleshooting and compliance monitoring.
- Experienced in automated workflow for software deployment by creating custom scripts either in **bash, PHP** and **ruby** in integration with **Jenkins**.
- Leveraged **APM** (Application Performance Monitoring) tools like **Datadog** and **AppDynamics** to identify performance bottlenecks and optimize AWS-hosted applications, achieving a **25% improvement** in system response times.
- Integrated security monitoring tools like **AWS Guard Duty** and **CloudTrail**, enhancing the detection and response to security threats, while ensuring compliance with industry regulations.
- Database management and administration using **Amazon RDS, PostgreSQL, Amazon DynamoDB** and **MSSQL** on **Docker** and Windows Server clusters.
- Conducted training sessions for users and team members to ensure they are proficient in utilizing Aws features and best practices.
- Working on hardware sizing, capacity management, **cluster management**, maintenance, performance monitoring and configuration of big data systems and applications for high volume throughput and processing.

- Designed, build and maintain near real-time big data applications and pipelines to process billions of records into and out of high-performance layers and data lakes, powering various data products and services.
- AWS network engineering including **VPC Peering**, **Transit Gateways, AWS Site-to-Site VPN, Transit VPCs, Hub VPCs, Palo Alto VM-Series, HA Proxy, Aviatrix Controller & Gateway**, **ELBs, NAT Gateways, Internet Gateways, OpenVPN and VPC endpoints/gateways/interfaces**.
- Utilized Kubernetes (EKS) cluster to perform rolling deployments by consuming **Docker** images pushed to **AWS ECR (Elastic Container Registry).**
- Involved in **JIRA** as defect tracking system and configure various workflows, customizations and plug-ins for JIRA bug/issue tracker, integrated Jenkins with **JIRA, GitHub**.
- Experience with deploying configuration management and CI/CD services such as (**Puppet, Ansible, PowerShell**, **Jenkins, Vagrant, Docker, JIRA, CloudFormation,** and **Elastic Beanstalk**).

<u>Environment</u>: AWS (EC2/Route53/AMIS/VPC/S3/RDS/DynamoDB/SNS/SQS, ECR, EKS), Ansible, Python, Splunk , Hadoop, Bamboo, SVN, RedHat Linux (4.x/5.x/ 6.x), CI/CD, Big data , GitHub, Maven, Nagios, OpenShift, Terraform, Power Shell, Bash Shell, Corn Jobs, SQL, Linux, AWS, Azure,GCP, Kubernetes, Docker, Cloud Formation, Jenkins, ELK Stack, Grafana, Prometheus, Python, Bash, Ipaas, Stackstorm, Hashicorp Vault, Zabbix, Graylog, Trendmicro.

Role: DevOps Automation Engineer **Aug 2022 – Jan 2024**
Client: IKIA, Action Netherland
Capgemini India Pvt Ltd
<u>Responsibilities:</u>

- Design and implement secure, scalable, and efficient **Cloud DevOps** platforms by leveraging industry standards, best practices, and Agile methodologies to support continuous integration and delivery **(CI/CD)** pipelines.
- Automate end-to-end CI/CD pipelines using tools like **Jenkins, Azure DevOps, GitHub Actions**, and **AWS CodePipeline** to enable rapid and reliable application deployments across development, testing, and production environments.
- Administer and optimize **Azure IaaS/PaaS services**, including **Azure Virtual Machines**, **VNET, Azure DevOps**, **SQL** Databases, Storage, **Azure Active Directory**, and Azure Search, ensuring high availability and performance.
- Manage and configure AWS services such as **ECR, Fargate, EKS, ECS, Lambda**, and **CloudFormation** to build, deploy, and maintain scalable and resilient cloud infrastructure.
- Develop and maintain Infrastructure as Code (IaC) solutions using **Terraform** (in collaboration with the **HashiCorp** team) and **AWS CloudFormation** (YAML templates) to automate the provisioning, configuration, and management of cloud resources.
- Containerize applications using **Docker** and orchestrate them using **Kubernetes** (**EKS, ECS**, and open-source **Kubernetes**) to ensure efficient deployment, scaling, and management of microservices.
- Troubleshoot and resolve issues with **AWS Elasticsearch Service** and other cloud services, optimizing performance and ensuring seamless operation of critical systems.
- Implement comprehensive monitoring and logging solutions using **ELK Stack** (**Elasticsearch, Logstash, Kibana**), **Grafana**, and **Prometheus** to provide real-time visibility into system performance, resource utilization, and error rates.
- Write and maintain automation scripts in **Python, Bash, Ruby, Java**, and Go to streamline repetitive tasks, improve operational efficiency, and reduce manual intervention.
- Configure and manage hybrid **Kubernetes** clusters, ensuring compliance with Kubernetes security standards, performing code coverage reviews, and conducting regular security scans.
- Develop and deploy serverless applications using **AWS Lambda functions,** enabling event-driven automation and reducing infrastructure management overhead.
- Automated **UiPath operations** using **PowerShell** through API calls and developed a **Python Tkinter** GUI to execute PowerShell commands seamlessly, reducing manual effort and improving operational efficiency.

- Built Lambda-based automation for monitoring **Glue jobs, RAS URLs, S3**, and **SQL long-running queries**, with metrics pushed to **CloudWatch** using **PutMetric**, enabling proactive issue detection and resolution.
- Triggered **Lambda functions** based on event patterns using **Event Scheduler** and **Event Bus**, executing specific code stored in **S3 buckets** to automate dynamic workflows and improve operational efficiency.
- Worked with **Azure Functions** and **GCP Functions** to automate various tasks, such as data processing, event-driven workflows, and backend integrations, improving scalability and reducing operational overhead.
- Led implementation of **Azure Active Directory** for single sign-on and Authentication for Web Applications. Also configured Azure Role-based Access Control (**RBAC**) to segregate duties within our team and grant only the amount of access to users that they need to perform their jobs based on Roles defined.
- Created and deployed VMs on the **Microsoft cloud service Azure**, created and managed the virtual networks to connect all the servers and designed Terraform for **Aws/Azure** platform.
- Developed and maintained Infrastructure as Code templates using **Terraform**, enabling consistent and automated provisioning of resources.
- Managed the enrollment of devices into **Microsoft Intune**, ensuring a smooth onboarding process for both corporate-owned and bring-your-own-device (BYOD) scenarios.
- Provides **RESTful APIs** for programmatic interaction with **Splunk**, allowing users to automate tasks, extract data, and integrate with other systems.
- Deployment Manager or **Terraform** to define, provision, and manage infrastructure as code, ensuring consistency and version control in **GCP environments**.
- Implemented monitoring solutions to track Couchbase performance metrics. Develop automated scripts for proactive optimization based on monitoring data.
- Implemented and optimized CI/CD pipelines using **Azure DevOps** and **GitHub actions**, reducing deployment time by X%.
- **GCP** supports **Kubernetes**, an open-source container orchestration platform, for deploying, managing, and scaling containerized applications.
- Integrated Golang applications with various databases (e.g, **PostgreSQL, MySQL, MongoDB**) using appropriate drivers and libraries, optimizing database interactions for performance and reliability.
- Integrated **Jenkins** pipeline jobs with the following build management and version control tools: **GitHub, Ansible, Chef, Rancher, Dockers, Terraform Cloud** and AWS to automated workflow.
- Designed and implemented highly available and fault-tolerant architectures on Azure, utilizing services like **Azure Load Balancer** and Traffic Manager.
- Demonstrated expertise in containerization technologies such as **Docker** and **Kubernetes**, enabling efficient deployment and management of applications.
- Utilized **Splunk** to monitor logs for detailed analysis and troubleshooting in Azure environments.
- Implemented monitoring and alerting solutions using **Prometheus** and **Grafana** to proactively identify and resolve issues in Azure environments.
- **WebAPI** are platform-independent and can be consumed by clients built with different technologies, including web browsers, mobile apps, and desktop applications.
- **WebAPI** supports various authentication and authorization mechanisms such as **API keys, OAuth**, and **JWT** (JSON Web Tokens) to control access to resources and ensure security.
- Created inventory in **Ansible** for automating CD & developed Ansible playbooks and Roles using YAML scripting.
- Worked on creation of **Docker** images on top of **micro services** and deployed on **Azure Kubernetes services**.
- Worked on Kubernetes cluster creation and creation of Deployments, services, **RBAC** and **Ingress**.

- Configured Service Now to receive any instant notifications of any configuration changes in cloud environment by orchestrating through Logic Apps.
- AWS Foundation including Cost Management, **AWS Budgets, AWS Trusted Advisor**, and other cost optimization tools to monitor, analyze, and control cloud spending, ensuring efficient resource utilization and cost-effective operations.
- DevOps practices involve continuous deployment and delivery. Configuring Vertex solutions to seamlessly integrate with **CI/CD pipelines** is essential for maintaining a smooth development and release process.
- Orchestrated the provisioning and configuration of Kubernetes clusters, utilizing tools like kops, kubespray, or cloud-managed **Kubernetes services** (e.g, **EKS, AKS, GKE**).
- Managing configurations and changes to Vertex implementations through version control systems to maintain traceability and facilitate collaboration among team members.
- Experience in setting up Baselines, Branching, Merging and Automation Processes using **Shell, Python** and **Bash Scripts**.

Environment: Azure, AAD, Azure DevOps, GCP, Splunk, Vertex, Couchbase, Terraform, Kubernetes (K8, AKS), Networking, OpenShift, Docker, Ansible, WEBAPI, Kafka, Prometheus, Grafana, Bash, Python, Linux, Jira, Bitbucket, CI/CD, GitHub, Apache Tomcat, ARM, Virtualization, CRON, AWS (EC2, VPC, ALB, Lambda, ECS, EKS, RDS, DynamoDB, Route53), Jenkins, Git, CloudFormation, PowerShell, ServiceNow, UiPath, Cron.

Role: SRE/DevOps Engineer **May 2021 – Apr 2022**
Client: WiserAdvisor, USA

Responsibilities:

- Created and configured **AWS EC2** instances using preconfigured templates such as **AMI, RHEL, Centos, Ubuntu** as well as used corporate based VM images which include complete packages to run build and test in those **EC2 Instances**.
- Managed and optimized **AWS services** (EC2, VPC, IAM, S3, CloudFront, RDS, ElastiCache, Lambda, DynamoDB, Kinesis, CloudWatch) and **Azure services** (Virtual Machines, Azure Functions) to ensure high performance, scalability, and reliability of the **WiserAdvisor** e-commerce platform.
- Designed and implemented CI/CD pipelines using **Docker** for containerization, **Kubernetes** for orchestration, and **GitLab CI/CD** for automation, enabling rapid and reliable deployment of application updates.
- Automated infrastructure provisioning and configuration management using **Terraform** (Infrastructure as Code) and **Ansible**, reducing manual intervention and ensuring consistency across **AWS** and **Azure** environments.
- Integrated Slack with monitoring tools like **Prometheus, Grafana, New Relic, Datadog**, and **AWS CloudWatch** to provide real-time alerts for system performance, error rates, and latency, enabling proactive issue resolution.
- Architected and deployed a highly scalable and fault-tolerant multi-cloud infrastructure using **AWS EC2 Auto Scaling**, **Elastic Load Balancers (ELB)**, **RDS** for **MySQL**, **ElastiCache** for Redis, **Azure Virtual Machines**, and **Azure Functions** to handle peak traffic and ensure high availability.
- Achieved **AWS Cloud Practitioner** certification and applied AWS Well-Architected Framework principles to design cost-effective, secure, and scalable cloud solutions.
- Monitored and optimized the e-commerce platform using **New Relic APM, Datadog, AWS CloudWatch**, and ELK Stack (**Elasticsearch, Logstash, Kibana**) to track performance metrics, identify bottlenecks, and improve response times.
- Collaborated with cross-functional teams to implement a microservices architecture, **RESTful APIs**, and message queues (**SQS**) for seamless integration of payment gateways, recommendation engines, and inventory management systems.

- Enhanced platform scalability and reliability by leveraging **AWS Lambda** for serverless computing, **Azure Functions** for event-driven processing, **DynamoDB** for high-speed data storage, and **Kinesis Data Streams** for real-time analytics on customer behavior and sales trends.
- Developed intuitive, responsive, and visually appealing user interfaces using **HTML, CSS**, and **JavaScript** to improve the overall user experience of the **WiserAdvisor platform**.
- Automated the build, test, and deployment processes using **Jenkins**, **AWS CodePipeline, Azure DevOps, Python**, **PowerShell, Docker**, and **Kubernetes**, ensuring faster and error-free releases.
- Configured and managed **Kubernetes** clusters for container orchestration, including workload scheduling, scaling, and self-healing capabilities across **AWS** and **Azure environments**.
- Designed and maintained **RESTful APIs** to facilitate seamless communication between various components of the e-commerce platform, such as **payment gateways** and recommendation engines.
- Conducted performance tuning and troubleshooting using **New Relic**, **Datadog**, **Prometheus**, **Grafana**, and **AWS CloudWatch** to ensure optimal system performance and reliability.
- Developed and maintained Infrastructure as Code (IaC) scripts using **Terraform** to automate the provisioning and management of cloud resources across **AWS** and **Azure**, ensuring consistency and repeatability.
- Ensured high availability and disaster recovery by designing and implementing robust **multi-cloud architectures**, including **AWS Backup**, **Asigra backup solutions**, multi-region deployments, and automated DR (**Disaster Recovery**) processes.
- Extensive use of **Elastic Load Balancing** mechanism with **Auto Scaling** feature to scale the capacity of **EC2 Instances** across multiple availability zones in a region to distribute incoming high traffic for the application.
- Develop and maintain automated deployment scripts using tools like **Ansible, Puppet**, or **Chef** to provision and configure Couchbase clusters consistently.
- Created, tested and deployed an End-to-End CI/CD pipeline for various applications using **Jenkins** as the integration server for **Dev, QA, Staging, UAT** and **Prod** Environments with Agile methodology.
- Created **Terraform** modules to create instances in **AWS** & automated process of creation of resources is AWS using **terraform**.
- Implemented and manage app protection policies in Intune to secure and control access to corporate data within mobile applications.
- Established proactive incident management processes using **Azure Monitor** and **Azure Application Insights**, resulting in a **20% reduction** in mean time to resolution **(MTTR)** for critical healthcare-related incidents.
- Implemented **Azure DevOps** pipelines for continuous integration and **continuous deployment** (CI/CD), streamlining application deployment and achieving a **30% reduction** in deployment time.
- Conducted regular performance analysis and optimization of **Azure services**, resulting in a **25% improvement** in system response times and a 15% reduction in resource costs through efficient resource scaling and utilization analysis.
- MVC frameworks handle **HTTP requests** by routing them to the appropriate Controller actions based on URL patterns and HTTP methods (**GET, POST, PUT**, and **DELETE**). Controllers then invoke the corresponding Model operations and select the appropriate View to render.
- Actively participated in on-call rotations, ensuring **24/7 support** for **Azure infrastructure** and contributing to the successful resolution of critical healthcare incidents with minimal impact on patient care.
- Worked with Site Reliability Engineer to implement **Data dog** system metrics, analytics, and dashboards.
- Built a deployment pipeline for deploying tagged versions of applications to **AWS beanstalk** using **Jenkins CI**.
- Used **Ansible** server to manage and configure nodes. Managed **Ansible Playbooks** with **Ansible roles**. Used file module in **Ansible playbook** to copy and remove files on remote systems.

Environment: Ansible, Docker ,Kubernetes, MVC , Finops Python, Bash, CI/CD, GitHub, WEBAPI , OpenShift, Splunk, Sumo Logic, New Relic, Service now, AWS (EC2, VPC, IAM, S3, CloudFront, RDS, ElastiCache, Lambda, DynamoDB, Kinesis, CloudWatch, Backup), Azure (Virtual Machines, Azure Functions, Azure DevOps), GitLab CI/CD, Terraform, Ansible, Python, Bash, Prometheus, Grafana, New Relic, Datadog, ELK Stack (Elasticsearch, Logstash, Kibana), Helm, ArgoCD, Slack, Jenkins, Istio, SQS, RESTful APIs, Microservices Architecture, Blue-Green Deployments, Canary Releases, HTML, CSS, JavaScript, PowerShell, Asigra, Nginx.

Role: Aws Cloud Engineer **Nov 2017—Apr 2021**
Client: E-Procurement (ITC)

Responsibilities:

- Installation, configuration and upgrade of **RHEL** 4/5/6 (and some 7), **Ubuntu**, **Oracle Enterprise Linux** 5/6/7 operating systems, automated infrastructure deployment using **AWS CloudFormation**, reducing manual effort and ensuring consistent environments across development, staging, and production.
- Automated the deployment of cloud infrastructure for the e-procurement buying application using **AWS CloudFormation**, reducing manual effort and ensuring consistent environments across development, staging, and production.
- Designed and implemented **IAM roles**, policies, and security groups to enforce granular access controls, ensuring secure access to the e-procurement application hosted on **AWS**.
- Configured **S3 lifecycle** policies and automated **RDS snapshots** to ensure data durability and disaster recovery readiness for the e-procurement platform.
- Monitored the performance of the e-procurement application using **AWS CloudWatch**, setting up custom alarms for key metrics such as **CPU utilization**, **disk space**, and application response times.
- Automated the scaling of **EC2** instances using **AWS Auto Scaling** to handle fluctuating workloads during peak procurement cycles, ensuring optimal performance and cost-efficiency.
- Utilized **AWS Cost Explorer** to analyze cloud spending patterns and implemented cost-saving measures, such as stopping unused **EC2 instances** and purchasing reserved instances.
- Improved the **CI/CD pipeline** for the e-procurement application using **AWS CodePipeline** and **CodeDeploy**, enabling faster and more reliable deployments with minimal downtime.
- Containerized key components of the e-procurement application using **Docker** and orchestrated them with **Amazon ECS**, improving scalability and resource utilization.
- Automated routine operational tasks, such as log rotation, backup management, and instance health checks, using **Python** and **Bash scripts**, reducing manual intervention.
- Designed and configured a secure **VPC** environment with **subnets, NACLs**, and security groups to isolate and protect the e-procurement application's infrastructure.
- Implemented **Amazon CloudFront** as a content delivery network **(CDN)** to improve the performance and reduce latency for global users of the e-procurement application.
- Automated the provisioning and management of **RDS databases**, including the setup of read replicas to handle high read traffic during procurement events.
- Managed and version-controlled the e-procurement application's cloud resources using **Terraform** as part of an Infrastructure as Code (IaC) approach.
- Integrated the **ELK Stack** (**Elasticsearch, Logstash, Kibana**) for centralized log management and analysis, enabling faster troubleshooting and performance optimization.

- Automated the deployment of application updates using **GitLab CI/CD**, reducing manual intervention and ensuring consistent deployments across environments.
- Configured **S3 bucket policies**, versioning, and encryption to securely store procurement-related documents and data, ensuring compliance with data protection standards.
- Collaborated with development teams to optimize the e-procurement application's architecture for high availability and fault tolerance on **AWS**, leveraging services like **ELB** and **Multi-AZ RDS**.
- Automated the creation of **Amazon Machine Images (AMIs)** for EC2 instances to ensure quick recovery and deployment during failures or scaling events.
- Developed **AWS Lambda functions** to automate event-driven tasks, such as triggering notifications for procurement approvals or sending system alerts for critical events.
- Documented and maintained runbooks and operational guides for the e-procurement application, ensuring smooth knowledge transfer and operational continuity for the team.
- Automated installation of **Linux servers** using Kickstart and Installation of **Solaris using Flash**, live upgrades and **Jumpstart Servers**.
- Administered **Linux-based servers**, ensuring optimal performance, security, and troubleshooting of issues across various distributions.
- Developed **Bash scripts** for automating routine tasks, streamlining processes, and ensuring consistency in Linux-based environments.
- Integrated Couchbase into **CI/CD pipelines**, automating testing, deployment, and rollback processes to streamline development workflows.
- Diagnosis and maintenance of hardware issues involving single and multiple point failures.
- Designed and developed **Talend ETL jobs** to **extract, transform**, and **load data** from various sources into data warehouses. Monitored and optimized **Talend jobs** to ensure efficient data processing and timely delivery.
- Configured **RDS logging** to capture detailed database activity logs, enabling better troubleshooting and performance tuning. Integrated RDS logs with **CloudWatch** for centralized monitoring and alerting
- Managed the uptime and maintenance of underlying servers, ensuring high availability and performance. Scheduled regular maintenance windows for updates, patches, and backups to minimize downtime
- Set up monitoring for **AWS Glue jobs** using **CloudWatch** to track job execution, success rates, and performance metrics. Automated alerts for job failures or delays to ensure timely resolution.
- Managed **DynamoDB tables**, including provisioning, scaling, and optimizing read/write capacity. Implemented **DynamoDB Streams** for real-time data processing and integrated with **Lambda** for event-driven workflows.
- Scheduled and orchestrated **Talend ETL** jobs using Talend Administration Center and integrated them with external schedulers like **Apache Airflow** or **cron jobs**. This ensured timely execution of data pipelines and seamless data flow across systems
- Optimized **Talend ETL** jobs by fine-tuning **SQL queries**, leveraging parallel processing, and reducing data redundancy. This improved job execution times by 30% and enhanced overall data processing efficiency.

Environment: RHEL, Ubuntu, Oracle Enterprise Linux5/6/7, Couchbase, API, OpenShift, GitHub, VMWare, Apache Tomcat, Shell **,** AWS (EC2, S3, VPC, IAM, CloudFront, RDS, CloudWatch, Lambda, CodePipeline, CodeDeploy, ECS, Auto Scaling), Azure (Virtual Machines, Blob Storage, Azure AD, Azure Monitor, Azure DevOps), Terraform, GitLab CI/CD, Docker, Kubernetes, Jenkins, Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Python, Bash, Slack, Jira, Confluence, Talend, DynamoDB, SQL, Redshift.